Quiz 7

1 Grading Scheme

- Question 1 is for 3 points
- Question 2 is for 3 points
- question 3 is for 3 points

Total grade is out of 6

2 Solutions

1. Suppose that the statement is false. Let X be the set of natural numbers n such that $n > 2^n$. Since $\mathbb{N}$ is well ordered, there exist a smallest element in X, say c. Because $0 < 2^0 = 1$, $c > 0$. Then we know that $c-1 \in \mathbb{N}$ and $(c-1) \notin X$ (by minimality of c). That is, $c-1 \leq 2^{c-1}$. Then we have that,

$$c \leq 2^{c-1} + 1 = 2^{c-1} + 2^0 \leq 2^{c-1} + 2^{c-1} == 2 \ \times 2^{c-1} = 2^c$$

   That is, $c \leq 2^c < c$, a contradiction. Hence, $X = \emptyset$ and for all $n \in \mathbb{N}$, $n < 2^n$.

2. Base case: n = 0, $0 = f_0 = f_2 - 1 = 1 - 1 = 0$.

   Induction hypothesis: Let $k \geq 0$. Suppose that for all $n \leq k$, $f_0 + f_1 + \ldots f_k = f_{k+2} - 1$.

   Induction step:

$$\begin{aligned} f_0 + f_1 + \ldots f_{k+1} &= (f_0 + f_1 + \ldots f_k) + f_{k+1} \\ &= f_{k+2} - 1 + f_{k+1} \\ &= (f_{k+1} + f_{k+2}) - 1 \\ &= f_{k+3} - 1 \\ &= f_{(k+1)+2} - 1 \end{aligned}$$

3. $S = 0, 2, 7, 15, 26, 40 \ldots$

   Solution 1

   Because the difference $s_i - s_{i-1}$ forms an arithmetic progression, we know that the polynomial $p(x)$ for S is quadratic. As $p(0) = 0$, the polynomial is of the form $p(x) = x(ax + b)$. Then we have

$$p(1) = a + b = 2$$

$$p(2) = 4a + 2b = 7$$

Solving for a and b we get a = 3/2 and b = 1 / 2. Hence, $p(x) = x(3x+1)/2$.

Solution 2

$$\begin{aligned} p(x) &= a_0\binom{x}{0} + (\Delta a_0)\binom{x}{1} + (\Delta^2 a_0)\binom{x}{2} + \dots \\ &= 0 + 2x + 3x(x-1)/2 \\ &= x(3x+1)/2 \end{aligned}$$

3 Common error

- In Question 1, need to show that $c - 1 \in \mathbb{N}$ by showing that $c \neq 0$.
- Please prove the base case and write the Induction hypothesis clearly.
- As a general remark, in induction proofs, start with k + 1 case and reduce it to an instance of k and use the induction hypothesis.
- Most answers were not polynomial in Question 3.
- An easy way to guess the degree of the polynomial is Question 3 is to check the largest k such that $\Delta^k(a_i)$ is nonzero.